EdgeCloud: Evidence Budgeting for Edge AI under Degraded Connectivity: A Validation Study of Adaptivity, Admissibility and Adversarial Metadata on Synthetic and Real Disaster Data

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***Abstract*—Edge AI on degraded field networks must decide what evidence to move and when to stop waiting. We formulate this as evidence budgeting across concurrent incident windows: a multiple-choice knapsack over a shared byte budget whose latency term prices transfer time together with the forecast delay of late evidence, so that waiting is an explicit decision with a closed-form threshold; evidence is fused only when its location, time and hazard tag are admissible. The formulation is compared with budget-matched simple rules that receive the same information, across nine synthetic conditions at twenty seeds and a reduced CrisisMMD workload at five. A derived delay threshold cut 95th-percentile latency from 1,562.3 s to 25.7 s with the error difference spanning zero; the admissibility filter rather than the optimizer keeps misaligned reports out; forged metadata erases its protection; a reused conformal threshold drops coverage to 0.745 against a nominal 0.90; an emulated TCP path confirms the latency model's byte term. On real data the optimizer leads the fixed rule in 15 of 21 event-split comparisons on the damage label, 14 of 21 once class imbalance is handled and 11 of 21 on the annotators' own label; a held-out contribution term cuts its errors from 447.0 to 386.2. The formulation earns its place by naming the quantities that matter and supplying a threshold, never by the solver.**

***Index Terms*—edge AI, degraded connectivity, multimodal fusion, evidence admissibility, conformal prediction, adversarial metadata, disaster data**

I. Introduction

A disaster produces imagery, field photographs, public reports and sensor alarms inside the same hour, almost none of it arriving together, over the network the event has just damaged. Three decisions are coupled: which evidence deserves the remaining bandwidth, whether a late observation is worth waiting for and whether a report belongs to this incident at all. This paper formulates them jointly, solves the program exactly and asks whether the formulation earns its complexity against simple rules that receive the same information (delay forecast, incident metadata, byte budget), since only an information-matched rule separates the value of a method from the value of what it is fed.

The answer is not the one we set out to demonstrate, so what a reader gains is stated at the outset: a closed-form delay threshold (6) that the simple rules do not supply, collapsed to a constant (7) that needs no solver; a held-out contribution term that repairs the value function where a modality carries no signal for the label (Section VIII-E); a recipe for deriving three candidate rules from the same quantities and choosing among them on held-out events; evidence about which assumptions fail (conformal coverage across conditions and events, the byte model, the dependence of every real-data ranking on the labelling); and two failure modes for contested deployments, admissibility collapsing with forging capability and the exact knapsack losing far more than a greedy rule without coordination. The system is analyst decision support; it does not dispatch.

II. Related Work and Positioning

xBD, FloodNet, Sen1Floods11 and CrisisMMD define what can be learned from disaster imagery and text [1]–[4], where joint representations improve humanitarian-category prediction [5], not what should be transmitted when the link degrades. Edge computing [6], [7], collaborative inference [8], communication-efficient edge AI [9], adaptive multimodal transmission [10], [11] and AVERY's intent-driven split computing [12] each serve one inference pipeline; none allocates one budget across concurrent incidents, prices waiting for late evidence or checks that evidence belongs to the incident being decided. Incomplete-modality learning [13], calibration [14], selective prediction [15] and split conformal prediction [16], [17] supply the uncertainty machinery; coverage does not survive distribution shift [18], so we calibrate per condition

and measure what reuse costs. No component is new and disaster-chain decision systems [19] and edge–cloud evidential fusion [20] predate this work; what is distinctive is that the alternatives receive the same information and the outcome is reported even when it goes against the proposed method.

III. System Model and Problem Formulation

A. Incident Windows and Admissible Evidence

An incident window i has a grid cell c_i, a decision time t_i and a hazard tag h_i; evidence item m carries its own and is admissible only when all three are compatible:

$$\mathrm{adm}_{im} = \mathbf{1}\big[\|c_{im} - c_i\|_\infty \le 1\big]\,\mathbf{1}\big[|t_{im} - t_i| \le \tau_m^{\mathrm{adm}}\big]\,\mathbf{1}\big[h_{im} = h_i\big]. \quad (1)$$

An inadmissible item may only be held. τ_m^{adm} is 7,200 s for imagery, 1,800 s for text and 600 s for sensor data; the cell tolerance is one grid step.

B. Actions, Objective and Why This Is a Knapsack

For each admissible item the policy chooses $a \in \mathcal{A} = \{\mathrm{hold}, \mathrm{summary}, \mathrm{feature}, \mathrm{raw}\}$: a summary carries calibrated local probabilities and metadata, a feature a quantised embedding, the raw action the observation. The K windows of a slot share one budget KB and one link, so the slot policy solves

$$\begin{aligned} \max_{x} \; & \sum_{i \in s} \sum_{m} \sum_{a \in \mathcal{A}} x_{ima} \Big[v_{im}(a) - \lambda_b \tfrac{C_{im}(a)}{B} - \lambda_\ell L_{im}(a) \Big] \\ \text{s.t.} \; & \textstyle\sum_a x_{ima} = 1 \quad \forall (i,m), \quad \sum_{i,m,a} x_{ima}\, C_{im}(a) \le KB, \\ & x_{ima} \in \{0,1\}, \quad x_{im,\mathrm{hold}} = 1 \text{ if } \mathrm{adm}_{im} = 0, \end{aligned} \quad (2)$$

where latency counts transfer time on the shared link of rate R and, for an item not yet at the edge, its forecast arrival delay:

$$L_{im}(a) = \frac{C_{im}(a)}{R} + \hat{D}_{im}\,\mathbf{1}[(i,m)\ \text{pending}], \quad L_{im}(\mathrm{hold}) = 0. \quad (3)$$

Holding costs nothing; queueing on the shared link applies to every method. Each item contributes one action under one shared budget, so (2) is a multiple-choice knapsack [21]. Evidence value is

$$v_{im}(a) = \phi(a)\ \hat{s}_i\ r_{im}\ f_{im}\ \big(\beta_0 + \beta_u u_{im} + \beta_q q_{im} + \beta_n n_{im}\big), \quad (4)$$

with fidelity $\phi = (0, 0.34, 0.64, 1.0)$ over (hold, summary, feature, raw), $\beta_0 = 0.28$, $\beta_u = 0.34$, $\beta_q = 0.23$, $\beta_n = 0.35$ and $\lambda_b = 0.11$. Severity $\hat{s}_i$, reliability r_{im} and recency $f_{im} = \exp(-\mathrm{age}_{im}/\tau_m)$ (τ_m of 3,600, 900 and 300 s) enter multiplicatively; uncertainty $u_{im} = 1 - \max_y p_{im}(y)$, quality q_{im} and novelty n_{im}, the Jensen–Shannon divergence [22] from the mean of the admissible evidence present, enter as a weighted sum over a positive floor. The coefficients are configuration set by hand.

C. The Wait-or-Act Decision

Because a pending item carries its forecast delay inside $L_{im}(a)$, waiting is not a heuristic beside the allocator. When the slot budget is not binding, (2) separates by item and a non-hold action is selected for a pending item only when its net utility is positive:

$$v_{im}(a) - \lambda_b \frac{C_{im}(a)}{B} - \lambda_\ell \Big(\frac{C_{im}(a)}{R} + \hat{D}_{im} \Big) > 0. \quad (5)$$

Rearranging gives a threshold on the forecast delay,

$$\hat{D}_{im} < T_{im}(a) \;\equiv\; \frac{v_{im}(a) - \lambda_b\, C_{im}(a)/B}{\lambda_\ell} - \frac{C_{im}(a)}{R}, \quad (6)$$

and the policy waits when $\max_{a \ne \mathrm{hold}} T_{im}(a) > \hat{D}_{im}$; sweeping λ_ℓ traces an accuracy–latency frontier. Collapsing the threshold to a constant gives a rule that needs no optimizer: for the image-summary action,

$$T^* = \mathrm{median}\{\, T_{im}(\mathrm{summary}) : (i,m) \in \mathcal{W}_{\mathrm{cal}} \,\} \quad (7)$$

over calibration windows whose image is pending, fixed before testing; the rule waits only when $\hat{D}_{im} < T^*$. Across the twenty seeds T^* was 193–231 s.

D. Properties

Rounding byte costs up to multiples of $\delta = KB/2000$ and solving (2) by dynamic programming [21] keeps every allocation feasible for the true budget and optimal whenever an optimum leaves at least $3K\delta$ bytes unused, 0.9% of the slot budget at $K = 6$. Under exchangeability within an operating condition the conformal set of (8) contains the true class with probability at least $1 - \alpha$ [16]; no guarantee is claimed across conditions and Section VIII shows the premise failing within one. Severity routes a case to review; it never confirms one.

IV. Method

Each edge node runs a lightweight model calibrated by temperature scaling [14]; the cloud holds a feature model trained only on the transmitted embedding and a raw model on the full observation, so no representation is scored with information it did not carry. Selected posteriors are combined by a weighted logarithmic opinion pool [23] with normalised weights proportional to $W_m q_{im} r_{im} f_{im}$ (W of 0.40, 0.35 and 0.25 for imagery, text and sensor data). The fused posterior becomes a split-conformal set [24] with the least-ambiguous set-valued classifier score [16],

$$\hat{C}(x) = \{\, y : p_i(y) \ge \hat{q} \,\}, \quad (8)$$

$\hat{q}$ being the $\lfloor \alpha(n_{\mathrm{cal}} + 1) \rfloor$-th smallest true-class probability over calibration windows of the same condition. A singleton set with confidence at least 0.90 from two modalities is confirmed, any other singleton likely, a non-singleton set insufficient evidence; divergence of at least 0.15 marks conflict and severity of at least 0.70 with either escalates to review.

V. Evaluation Protocol

A synthetic generator creates three classes and image, text and sensor vectors for 12 events with 180 windows each; events D00–D05 train, D06–D07 calibrate and D08–D11 are held out. Byte sizes are assumptions that Section VIII finds wrong. Windows form slots of $K = 6$ sharing budget and link, served in order of estimated severity; under delayed imagery 70% of

TABLE I

LOW BANDWIDTH (256 KBPS, 4,000 BYTES PER WINDOW). ERRORS OUT OF 720, MEAN OF TWENTY SEEDS WITH 95% INTERVALS; EVERY COLUMN, THE QUEUE SHARE INCLUDED, IS FROM THAT RUN. *IGNORES THE BUDGET.

Method	Errors [95% CI]	Bytes	p95 s	Queue
Image only	71.5 [59.1, 83.9]	224	0.27	7%
Text only	75.7 [63.9, 87.4]	196	0.26	6%
Transmit all*	4.3 [2.9, 5.7]	423,502	84.64	33%
Confidence gate	7.0 [5.0, 9.0]	1,063	0.47	30%
Fixed compression	4.4 [3.0, 5.8]	3,840	0.95	32%
Fixed rule	5.2 [3.6, 6.7]	1,860	0.59	34%
Random within budget	22.6 [18.5, 26.6]	1,695	0.60	27%
Proposed	4.4 [3.2, 5.6]	3,604	0.92	32%

TABLE II

DELAYED IMAGERY. ERRORS OUT OF 720 WITH 95% INTERVALS, MEAN OF TWENTY SEEDS; MEDIAN AND 95TH-PERCENTILE MODELLED LATENCY; SHARE OF IMAGES SENT. *IGNORES THE BUDGET.

Method	Errors [95% CI]	Median s	p95 s	Image sent
Image only	71.5 [59.1, 83.9]	682.80	1,864.4	100%
Text only	75.7 [63.9, 87.4]	0.15	0.15	0%
Transmit all*	19.2 [15.5, 23.0]	688.69	1,869.5	100%
Confidence gate	25.6 [21.1, 30.0]	682.81	1,864.4	100%
Fixed compression	18.9 [14.9, 23.0]	682.84	1,864.4	100%
Fixed rule	23.4 [18.7, 28.1]	682.82	1,864.4	100%
Random within budget	32.1 [28.1, 36.2]	39.69	1,664.7	68%
Wait if uncertain	16.0 [12.3, 19.7]	42.56	1,562.3	53%
Derived threshold T^*	10.8 [8.6, 13.0]	0.18	25.7	31%
Proposed, $\lambda_\ell = 0.0004$	10.2 [8.3, 12.0]	0.19	364.9	37%

images arrive after a log-normal delay (median 900 s) and the policy sees only a forecast with multiplicative log-normal error ($\sigma = 0.4$). The base configuration has no MAC layer, routing or contention. Nine conditions: normal (10 Mbps, 360 kB per window), low bandwidth (256 kbps, 4,000 bytes), delayed imagery, intermittent outage, image missing, text missing, noisy text, contradictory evidence and misaligned evidence (30% of text items from other events, 25% of those with compatible metadata). Eleven methods: the eight of Table I, wait if uncertain, the derived threshold (7) and the fixed rule with admissibility (1); the fixed rule sends an image summary, raw text and raw sensor data. Each seed provides 720 test windows; we report means of twenty seeds with two-sided 95% t intervals and paired seed-level differences. Five integrity properties, among them that budget-compliant methods never exceed a slot budget and that the wait threshold comes from calibration events only, are tested rather than assumed. Section VI is confirmatory; the main protocol was not pre-registered, the contested-budget experiments were, with frozen protocol files whose SHA-256 digests are checked at run time; Sections VII and VIII were designed afterwards and are exploratory.

VI. CONFIRMATORY RESULTS ON THE SYNTHETIC BENCHMARK

A. When Bandwidth Is the Only Constraint, a Fixed Rule Is Enough

Under low bandwidth (Table I) the policy and the fixed rule made statistically indistinguishable numbers of errors, a paired difference of -0.8 $[-1.7, 0.2]$, while the policy spent 1,745 more bytes per window and added 0.33 s to tail latency. Wherever the budget alone binds (normal connectivity, an outage, missing imagery) the paired difference spans zero. Once the evidence degrades it does not: with text missing the policy is ahead by 2.75 errors $[1.03, 4.47]$, under noisy text by 1.30 $[0.13, 2.47]$ and under contradictory evidence by 2.50 $[0.80, 4.20]$. When no decision involves waiting, admissibility or degraded evidence, the optimizer adds cost rather than value.

B. Waiting for Late Evidence: The Threshold Matters, the Solver Does Not

Methods that always wait paid a median latency near 682.8 s (Table II); the policy sent 37% of images, holding its median at

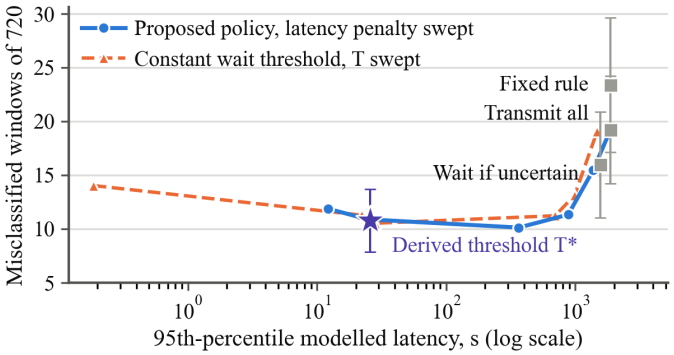


Fig. 1. Delayed imagery: errors against 95th-percentile modelled latency. The solid line traces the policy across latency-penalty values; the dashed line traces the constant-threshold rule across T. The star marks T^* derived from calibration events, not selected on test data. The sweep and the marked points are the twenty-seed run behind Table II.

0.19 s; removing its latency penalty cost 9.1 errors $[5.6, 12.5]$. The reduction does not require the optimizer: the derived-threshold rule sent 31% of images and made 10.8 errors at 25.7 s tail latency; relative to it the policy changed errors by -0.7 $[-1.5, 0.2]$ while adding 339.2 s $[314.4, 363.9]$ of tail latency through per-window adaptivity; Fig. 1 shows the two frontiers within one error of each other below 30 s. One caveat changes the conclusion: the policy's -13.3 errors $[-17.3, -9.2]$ against the fixed rule exist because the benchmark decays text and sensor evidence while a decision waits; with that decay removed the forecast-ignoring rules improve by 10.6 to 18.3 errors and the ordering reverses (fixed rule 5.2 against 9.7 for the policy and 10.7 for the threshold). Whether acting early improves accuracy depends on how fast real evidence goes stale; Section VIII-C tests that on a corpus with no decay.

C. Admissibility, Not Optimization, Keeps Foreign Reports Out

When 30% of text items came from other events, methods without the filter fused all of them; the proposed method fused the 24% whose metadata had been made compatible, rejected no aligned item and made 13.9 errors against 35.0 for the unfiltered fixed rule (-21.1 $[-26.2, -15.9]$) and 15.3 for the fixed rule with the same filter (-1.5 $[-2.7, -0.3]$). The

filter accounts for almost all of the gain; disabling it inside the proposed system added 16.2 errors [11.1, 21.3]. The test favours the filter by construction: zero false rejections follow from the generator, while real metadata carries positional error, clock skew and missing geotags.

D. Coverage Fails Across Conditions; the Review Queue Is the Cost

Calibrating per condition kept coverage between 0.889 and 0.899 against a nominal 0.90; reusing one threshold drops it to 0.794 under contradictory evidence and 0.745 with images missing (five-seed sweep), so a deployment that calibrates once at commissioning is miscalibrated exactly when the network degrades. Under low bandwidth about 30% of windows entered the review queue, 98% of them correctly classified; conformal abstention alone queued about 10% and caught every error, while the pre-specified conflict rule nearly tripled the queue for at most two additional caught errors per 720 windows, so the conflict threshold is a staffing decision. Severity escalation fired on 0.07% of windows against the 28% the generating severity would have, because the severity estimator is weak.

E. When the Shared Budget Binds (Pre-registered)

None of the nine conditions makes the shared budget bind, the only regime in which the knapsack can differ from a per-window rule. A frozen protocol set observation noise to 1.75, per-window budgets from 700 to 1,900 bytes at 256 kbps and ten seeds, against a shared-budget greedy rule that upgrades items in order of local uncertainty, with Holm adjustment [25] across the five budgets. The optimizer did not beat the competitor at any budget: at 700 bytes it was worse by +20.8 errors [15.4, 26.2] (Holm $p < 0.001$), because the value function prices a held modality at zero and the exact solver held a modality in 47% of windows to buy raw sensor records elsewhere; at the three largest budgets it led nominally by −4.0, −3.6 and −4.8 errors, none surviving adjustment (Holm p 0.075 to 0.219). A greedy solver on the same utility matched the exact one within one error at every budget; one difference, 1.0 error in 720 at B = 1,500, survived adjustment in the solver's favour ($p = 0.042$). A summary-floor variant, frozen as a second protocol on ten new seeds, removed the tight-budget failure without any difference surviving adjustment. Exact optimization does not repair a misspecified objective; it exploits it.

VII. Exploratory Analyses

These were designed after the confirmatory results; each inherits the perturbation stream of the condition it modifies. *Adversarial metadata.* Varying the forged fraction of misaligned reports collapses the filter almost linearly, worth 7.0 errors against the unfiltered rule's 20.2 with no forgery and nothing at full forgery, the admitted fraction tracking the forged fraction (24.1%, 48.4%, 72.4% against 25%, 50%, 75%); admissibility is a filter against accident, not a security control. *Packet loss.* Packet loss with retransmission leaves accuracy unmoved and widens the tail-latency gap between the fixed rule and the policy from 6× to 34× at 10% loss (0.14 s against 4.70 s). *Measured links.* Replaying LTE traces recorded for the Sprout study [26] and distributed with Mahimahi [27] against a constant link of equal mean capacity inflates tail latency by up to 8.7×; the ranking survives, the idea that the optimizer's extra bytes are cheap does not. *Emulator.* Every payload of the byte model was sent over TCP through Mahimahi link shells in 17 cells, 2,890 transfers, none failed. The byte term of (3) holds (the 420 kB record took 14.05 s on the 256 kbps link against 13.35 s modelled); the additive constant is one round trip short, slow start [28] sets the record's time on fast links (1.31 s against 0.40 s at 19 Mbps) and the loss term misses both ways, so every latency small against a round trip is a lower bound. *Coordination.* Dealing a slot's windows to C independent controllers costs the exact knapsack +20.0 errors [14.2, 25.8] at $B = 900$ against +3.9 [0.8, 7.0] for the greedy ranking; at $C = 6$ the policy makes 92.6 errors against 75.1.

VIII. A Reduced Comparison on Real Disaster Data

A. Workload and the Text-Pathway Diagnostic

CrisisMMD [4] supplies 3,526 labelled report–image pairs from seven disasters in 2017 with real text, photographs, hazard types, arrival times and payload sizes; the policies are unchanged and the split is by event. Both modalities are CLIP embeddings [29] reduced by principal components fitted on training events (eight for text, ten for images). The corpus has no sensor stream, no geolocation (the spatial term of (1) reduces to event identity), no severity independent of the label and no delayed imagery, so three of nine conditions and eight of eleven methods carry over. Five seeds vary model fitting only on the 968 test windows of the primary split (795 from Hurricane Irma, 173 from the Iraq–Iran earthquake), so the event list is also rotated through seven pre-specified 3/2/2 assignments. Before any budgeting decision the image representation reaches 0.566 accuracy against a majority baseline of 0.490 while the text representation returns the majority class for every record. A twenty-five-cell diagnostic over five text representations, three class handlings [30] and three classifiers, with its pass rule fixed in advance (clear the majority baseline by 0.02 accuracy and beat its macro-F1 in five of seven rotations), returns false in every rotation: the full embedding with balanced resampling raises macro-F1 from 0.219 to 0.452 while its accuracy stays below the baseline. The pathway carries class-discriminative information that accuracy on imbalanced classes hides yet predicts this label too poorly to clear the bar; the cause is class imbalance, not the reduction.

B. Results

Against the fixed rule the synthetic ranking does not carry over (Table III): the optimizer wins under normal connectivity by 17.2 errors [15.8, 18.6] and under misaligned evidence by 11.2 [7.8, 14.6] and loses under low bandwidth by 1.4 [0.7, 2.1]. Across the seven rotations it is ahead in 15 of 21 comparisons and under normal connectivity in all seven, but the low-bandwidth loss occurs in 2 rotations and the misaligned win in 3, so which way a single condition falls

TABLE III
ALL THREE REAL-DATA CONDITIONS. ERRORS OUT OF 968; THE MAJORITY-CLASS BASELINE IS 494. BYTES AND TAIL LATENCY ARE FOR LOW BANDWIDTH. *IGNORES THE BUDGET.

Method	Normal	Low bw.	Misaligned	Bytes	p95 s
Image only	420.0	420.0	420.0	224	0.27
Transmit all*	435.8	435.6	447.4	129,728	30.71
Fixed rule	464.2	465.4	473.4	342	0.29
Confidence gate	437.0	465.8	474.4	1,390	0.56
Random within budget	462.0	465.8	474.6	1,212	0.52
Greedy by uncertainty	466.8	466.8	476.0	2,166	0.63
Greedy, proposed utility	447.0	466.8	462.2	1,896	0.63
Proposed (exact knapsack)	447.0	466.8	462.2	1,896	0.63
Fixed compression	483.0	483.0	486.0	3,072	0.80
Text only	494.0	494.0	494.0	196	0.26

depends on which disasters land in the test set. Exact solver and greedy heuristic on the same utility are identical in every cell, while a greedy ranking by local uncertainty makes 466.8 errors against 447.0: the utility works, the solver does not. Image-only beats everything at 420.0 errors, because text-only scores the majority count of 494.0 and everything fusing text pays for it; no term of (4) asks whether a modality has ever predicted this label. Coverage falls to 0.816 under normal connectivity and 0.745 under low bandwidth, because calibration and test windows come from different disasters. The byte model is contradicted: real images run from 1,763 to 1,719,070 bytes (median 103,142.5) and every report (39 to 181 bytes) is smaller than the smallest summary.

C. The Constant Threshold on Real Data

With an arrival delay constructed from the synthetic log-normal family under a frozen protocol, T^* computed on calibration events lands between 164.0 and 186.8 s and holds 95th-percentile latency at 25.1 to 26.8 s against 1,739.2 to 1,914.8 s for the always-wait rules in all seven rotations, a 67- to 74-fold reduction. Its accuracy gain does not carry over: it costs 35.0 errors on the primary split (486.0 against 451.0), ranges from 16.0 fewer to 36.2 more and meets the pre-specified rule in one rotation where five were required. The corpus does not decay evidence while a decision waits, so latency is bought with accuracy rather than alongside it.

D. Class Imbalance and a Label the Paper Did Not Choose

With balanced resampling [30] applied to every per-modality model the optimizer stays ahead of the fixed rule in 14 of 21 comparisons and the primary-split gap widens from -17.2 to -112.6 errors (optimizer 447.0 to 385.6, transmit-all 435.8 to 377.8) while every single-modality rule worsens (image-only 420.0 to 466.6), so the rule that beats everything in Table III does so partly because the imbalance was left in place. On the annotators' own humanitarian label (three most frequent categories, 2,763 of 3,526 records) the optimizer is ahead in 11 of 21 comparisons and in 3 of 7 rotations under normal connectivity rather than all 7, with class handling as well in 8 of 21: the direction of the real-data result is a property of the label as much as of the method.

E. A Held-Out Contribution Term and the Published Tasks

The value function gains the term it lacked: each modality's balanced accuracy above chance on the calibration events only, relative to the best modality, multiplied into (4). On the damage label text's estimate is 0.000 at every seed, so the policy stops sending it and its errors fall from 447.0 to 386.2 under normal connectivity, 466.8 to 431.0 under low bandwidth and 462.2 to 431.0 under misaligned evidence, below transmit-all everywhere; across the rotations 8 of 21 cells improve, 9 are unchanged and 4 worsen where the calibration disasters misjudged text; on the synthetic benchmark the term moves 4 of 180 cells by one error. On the two published CrisisMMD tasks of [5] on their agreed-label splits, with the pipeline generalised to K classes, transmit-all reaches 91.4% and 86.3% against the published 84.4% and 78.4%, the text pathway carries signal (82.7% against the published 80.8%) and the damage-label pattern repeats: the optimizer ahead of the fixed rule under normal connectivity and misaligned evidence (130.6 against 142.2 on the informative task) and behind under low bandwidth (148.0 against 142.4).

IX. LIMITATIONS AND THREATS TO VALIDITY

The synthetic data are well separated, which explains error rates below 1% for the stronger methods; byte sizes, delay distributions, forecast errors and the rate at which evidence ages are assumptions on which the waiting result depends (Section VI-B). On real data the link, budgets, loss model, slot structure, sensor modality and severity stay synthetic or fixed, the arrival delay used to test the threshold is constructed and the contribution term fails where the calibration events differ from the test events. There is no hardware testbed, no radio and no energy measurement; the substitutes, a real TCP stack under emulated links and replayed LTE traces recorded for an earlier study [26], do not stand in for a device, so four claims remain untested on hardware: every latency smaller than a round trip, which the emulator shows to be a lower bound; the loss term, which the emulator contradicts at 10% loss; the compute cost, reported as workstation wall-clock time and peak allocation with no joules inferred; and the cost of the per-slot coordination the benchmark charges nothing for. A deployment on edge hardware over a degraded radio would be the test; it has not been run, so every statement here about edge deployment is bounded by simulation and emulation on a workstation. Policy coefficients were set by hand; the main protocol was not pre-registered; the budget grid and competitor of the contested test were the authors' choice; three of the eight paired error comparisons include zero and are treated as inconclusive, not as equivalence; the seven real-data rotations reuse the same events. Improved disaster outcomes cannot be claimed without prospective studies with emergency-management partners.

X. REPRODUCIBILITY, OVERSIGHT AND DATA AVAILABILITY

The package contains source, environment lock, configuration, splits, seeds, integrity tests, record-level predictions, provenance envelopes, the two frozen protocols and scripts

that regenerate every table and figure; reruns reproduce every output byte for byte on one architecture; re-deriving the real-data records with a later, unpinned encoder stack changed no cell of Table III by more than 1.0 error with every sign preserved. The package is maintained at https://github.com/bdata-arindam/Edge at version 2.6.18 with a `SHA256SUMS` digest for each of its 234 files. The archived release is `10.5281/zenodo.XXXXXXX`. CrisisMMD v2.0 was obtained from its authors under their terms and is not redistributed. The method is decision support under the NIST AI RMF lifecycle [31].

XI. Conclusion

We formulated disaster evidence transmission as a shared-budget knapsack across concurrent incidents in which waiting is an explicit decision with a delay threshold and admissibility is checked before any evidence is fused. The value of the formulation is not in its solver: the simple rules it generates match the optimizer wherever the budget alone binds, a pre-registered test found the optimizer worse than its competitor at the tightest budget, the admissibility filter's protection falls to nothing in proportion to forging capability and the conformal guarantee does not transfer across conditions. On real data the optimizer's lead over the fixed rule falls from 15 of 21 comparisons to 14 of 21 once class imbalance is handled and 11 of 21 on the annotators' own label; the text pathway's majority-class behaviour is a class-imbalance effect that the value function detects only once a held-out contribution term is added, cutting the optimizer's errors from 447.0 to 386.2; the delay threshold keeps a 67- to 74-fold tail-latency reduction in every split while its accuracy gain does not carry over. The formulation earns its place by naming the quantities that matter and supplying the threshold, never by the solver; which simple rule wins is a property of the workload, the class handling and the labelling, so it has to be derived and tested on held-out data labelled as the deployment labels it. Field value on edge hardware and radios remains to be established.

References

[1] R. Gupta et al., "Creating xBD: A dataset for assessing building damage from satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2019, pp. 10–17.

[2] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89644–89654, 2021, doi: 10.1109/ACCESS.2021.3090981.

[3] D. Bonafilia, B. Tellman, T. Anderson, and E. Issenberg, "Sen1Floods11: A georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2020, pp. 835–845, doi: 10.1109/CVPRW50498.2020.00113.

[4] F. Alam, F. Ofli, and M. Imran, "CrisisMMD: Multimodal Twitter datasets from natural disasters," in *Proc. 12th Int. AAAI Conf. Web Social Media (ICWSM)*, 2018.

[5] F. Ofli, F. Alam, and M. Imran, "Analysis of social media data using multimodal deep learning for disaster response," in *Proc. 17th Int. Conf. Inf. Syst. Crisis Response Manage. (ISCRAM)*, 2020.

[6] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, 2016.

[7] M. Satyanarayanan, "The emergence of edge computing," *Computer*, vol. 50, no. 1, pp. 30–39, 2017.

[8] Y. Kang et al., "Neurosurgeon: Collaborative intelligence between the cloud and mobile edge," in *Proc. 22nd ACM Int. Conf. Archit. Support Program. Lang. Oper. Syst. (ASPLOS)*, 2017, pp. 615–629, doi: 10.1145/3037697.3037698.

[9] Y. Shi, K. Yang, T. Jiang, J. Zhang, and K. B. Letaief, "Communication-efficient edge AI: Algorithms and systems," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2167–2191, 2020.

[10] Q. Guo, Z. Yang, Y. Hu, C. Zhao, and W. Ji, "DAT: Dual-aware adaptive transmission for efficient multimodal LLM inference in edge-cloud systems," 2026, arXiv:2604.05375.

[11] H. Chi et al., "CST: Collaborative selective transmission for communication-efficient multimodal edge inference," 2026, arXiv:2608.22115.

[12] R. Bhattacharjya et al., "AVERY: Intent-driven adaptive VLM split computing via embodied self-awareness for efficient disaster response systems," Nov. 2025, arXiv:2511.18151v1.

[13] Y. Chen, M. Zhao, and L. Bruzzone, "Incomplete multimodal learning for remote sensing data fusion," 2023, arXiv:2304.11381.

[14] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.

[15] Y. Geifman and R. El-Yaniv, "SelectiveNet: A deep neural network with an integrated reject option," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 2151–2159.

[16] M. Sadinle, J. Lei, and L. Wasserman, "Least ambiguous set-valued classifiers with bounded error levels," *J. Amer. Statist. Assoc.*, vol. 114, no. 525, pp. 223–234, 2019.

[17] Y. Romano, M. Sesia, and E. Candès, "Classification with valid and adaptive coverage," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 3581–3591.

[18] Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.

[19] "Methods, devices, media and equipment for disaster response decision making based on multimodal approaches," China Patent Appl. CN121981868A, May 5, 2026.

[20] "A method for edge-cloud collaborative multimodal conflict resolution based on deep evidence uncertainty," China Patent Appl. CN119337322A, Jan. 21, 2025.

[21] H. Kellerer, U. Pferschy, and D. Pisinger, *Knapsack Problems*. Berlin, Germany: Springer, 2004, doi: 10.1007/978-3-540-24777-7.

[22] J. Lin, "Divergence measures based on the Shannon entropy," *IEEE Trans. Inf. Theory*, vol. 37, no. 1, pp. 145–151, 1991, doi: 10.1109/18.61115.

[23] T. Heskes, "Selecting weighting factors in logarithmic opinion pools," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 10, 1998.

[24] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005, doi: 10.1007/b106715.

[25] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, pp. 65–70, 1979.

[26] K. Winstein, A. Sivaraman, and H. Balakrishnan, "Stochastic forecasts achieve high throughput and low delay over cellular networks," in *Proc. 10th USENIX Symp. Networked Syst. Design Implement. (NSDI)*, 2013, pp. 459–471.

[27] R. Netravali et al., "Mahimahi: Accurate record-and-replay for HTTP," in *Proc. USENIX Annu. Tech. Conf. (ATC)*, 2015, pp. 417–429.

[28] M. Allman, V. Paxson, and E. Blanton, "TCP congestion control," IETF RFC 5681, Sep. 2009, doi: 10.17487/RFC5681.

[29] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748–8763.

[30] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.

[31] NIST, "Artificial intelligence risk management framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.